

chapter 4. Stability of Stochastic Impulsive System with Time Delay.

$$\begin{cases} dx(t) = f(t, x(t))dt + g(t, x(t))dw(t), & t = T_k \\ \Delta x(t) = \mathcal{F}(t, x(t-)). & t = T_k \\ x_{t_0} = \phi(s), \quad s \in [-r, 0] \end{cases} \quad \begin{matrix} (4.1a) \\ (4.1b) \\ (4.1c) \end{matrix}$$

Main interest: (1) mean square (m.s) global asymptotic stability
(2) problem of stabilization by impulsive controller.

Similar to ODE. 4.1 Stability analysis by Lyapunov technique.
4.2 Stability analysis by comparison method.

Assumption A1:

$$\exists 0 < p_1 \leq p, \forall T_k \in \mathbb{R}_+ \quad x \in \mathcal{PF}([-r, 0]; V), D \subset \mathbb{R}^n, \text{ If } E[\|x_{T_k}^*\|^2] \leq p, \\ \Rightarrow E[\|x(T_k)\|^2] < p. \quad \left| \begin{array}{l} \text{solution be bounded after any impulsive effect as} \\ \text{long as it is bounded before impulsive moment.} \end{array} \right.$$

Assumption A2: $\forall k, T_{\sup} := \sup\{T_k - T_{k-1}\} < \infty, T_{\inf} := \inf\{T_k - T_{k-1}\} > 0.$

4.1. Lyapunov technique.

refer to page 8.

Theorem 4.1. $\exists a, c \in \underline{k}_c, b \in \underline{k}_v, p \in \mathcal{PF}(\mathbb{R}_+, \mathbb{R}_+)$. Let $V \in \mathcal{V}^{1,2}([-r, \infty) \times \mathbb{R}^n, \mathbb{R}_+)$. s.t: $\begin{matrix} \text{concave} & \text{convex} \end{matrix}$

(i) $\forall (t, \psi_{(0)}) \in [r, \infty) \times S(p).$

$$b\| \psi_{(0)} \|^2 \leq V(t, \psi_{(0)}) \leq a\| \psi_{(0)} \|^2.$$

(ii) $\forall t \neq T_k \in \mathbb{R}_+, \psi \in \mathcal{PF}([-r, \infty); S(p)).$

$$\begin{matrix} f'(x) \leq c f(x) \\ f'(x) \geq -c f(x) \end{matrix}$$

$$L V(t, \psi) \leq p(t) \leq c V(t, \psi_{(0)}) \quad (\text{a.s.})$$

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provided that $\bar{g}(V(t+s, \psi(s))) \leq V(t, \psi_{(0)})$ for some $s \in [-r, 0], \bar{g} \in \underline{k}_3.$

(iii) at any impulsive moment, $T_k \in \mathbb{T}, \psi \in \mathcal{PF}([-r, \infty); S(p)).$

stability condition with impulsive system.

$$V(T_k, \psi_{(0)}) + \mathcal{F}(T_k, \psi(T_k^-)) \leq \bar{g}(V(T_k, \psi_{(0)})) \quad (\text{a.s.})$$

With $\psi_{(0)} = \psi_{(0)}$, where $(T_k, \psi(T_k^-)) \in (\mathbb{R}_+ \times \mathcal{PF}([-r, 0]; S(p)));$

$$(iv) M_1 = \sup_{t \geq 0} \int_t^{t+T} 1 \, ds < \infty, T = \sup_{k \in \mathbb{N}} [T_k - T_{k-1}] < \infty, M_2 = \inf_{t \geq 0} \int_t^{t+T} \frac{ds}{\bar{g}(s)} > M.$$

Then, the trivial solution $x=0$ of (4.1) is uniformly asymptotically stable in the m.s. (iv) means impulsive change unstable $\Rightarrow$ stable.

2) proof. From condition ci) , $\exists \hat{b} \in K_V, \hat{a} \in K_C. \hat{b}(s) \leq b(s) \leq a(s) \leq \hat{a}(s), \forall s \in [0, \infty)$
 This implies $\hat{b}(\|\phi\|_r^2) \leq V(t, \phi(s)) \leq \hat{a}(\|\phi\|_r^2)$ (a.s.) ($\phi \in \mathcal{PF}([t, \infty); \mathbb{R}^n]$)

We split the proof into two steps:

Step A: We first prove uniform stability in the m.s., i.e., $\forall \varepsilon > 0, \forall t_0, \exists \delta > 0$.

s.t. $E[\|\phi\|_r^2] \leq \delta \Rightarrow E[\|x(t)\|^2] < \varepsilon, \forall t \geq t_0$.

Step B: $x \equiv 0$ is uniformly asymptotically stable in the m.s.

From Step A, we obtain that.

$$\forall \delta > 0, \exists \eta > 0. \text{ s.t. } E[\|\phi\|_r^2] \leq \eta \Rightarrow E[\|x(t)\|^2] \leq \delta.$$

We only need to find some $T(\delta, \eta) > 0$, $\eta > 0$, s.t. $\forall \delta > 0, \exists T(\delta, \eta) > 0$.
 $E[\|\phi\|_r^2] \leq \eta \Rightarrow E[\|x(t)\|^2] \leq \delta, \forall t \geq t_0 + T$.

• proof of step A: Take $0 < \varepsilon < \rho_1$, and $x(t)$ be a solution of (4.1) in $[t_0, t_0 + \beta]$.
 Choose $\delta < \hat{a}^{-1}(\hat{g}(b(\varepsilon))) \Rightarrow 0 < \delta < \varepsilon$ (since $0 < \hat{g}(b(\varepsilon)) < \rho_1$).

If claim was not true, $\exists t \in [t_0, t_0 + \beta]$ such that $E[\|x(t)\|^2] > \varepsilon$.

Then, define $\bar{t} = \inf \{ t \in [t_0, t_0 + \beta] \mid E[\|x(t)\|^2] > \varepsilon \}$.

clearly $E[\|x(t)\|^2] \leq E[\|\phi\|_r^2] \leq \delta < \varepsilon, \forall t \in [t_0 - r, t_0] \Rightarrow \bar{t} \in (t_0, t_0 + \beta)$.

and $E[\|x(t)\|^2] \leq \varepsilon \leq \rho_1, \forall t \in [t_0 - r, \bar{t}]$.

For $\bar{t}$, ① $\bar{t} \neq T_k \Rightarrow E[\|x(\bar{t})\|^2] = \varepsilon$ (by continuity of $x(t)$).

② $\bar{t} = T_k \xRightarrow[\text{(for some } k)]{\text{maybe}} E[\|x(\bar{t})\|^2] > \varepsilon, E[\|x(\bar{t}^+) \|] \leq \rho_1$ (by A1)

From Itô formula, we have.

$$V(t, x(t)) = V(s, x(s)) + \int_s^t LV(u, x(u)) du + \int_s^t V_x g dW.$$

$E(\cdot) \geq 0$.

Take the expectation, we get.

$$m(t) = E[V(t, x(t))] \leq E[V(s, x(s))] + E \int_s^t L v du.$$

$$\leq m(s) + E \int_s^t p(u) \leq (V(u, x(u))) du. \quad (\text{by Cii})$$

This implies

$$p^+ m(t) \leq p(t) E[V(t)] \leq p(t) m(t), \forall t \neq T_k \in [t_0, \bar{t}]$$

(by concave of c).

provided $m(t) \geq \hat{g}(c(\|m(t)\|_r))$. (*)

$\forall T_k \in (t_0, \bar{t}]$, by Ciii), we obtain $m(T_k) \leq \hat{g}(c(m(T_k)))$.

Let $t^* = \inf \{ t \in [t_0, \bar{t}] \mid m(t) \geq \hat{b}(\varepsilon) \}$.
 * ($t^* > t_0$) $m(t_0) \leq \hat{a}(c(\|\phi\|_r^2)) \leq \hat{a}(s) < \hat{a} < \hat{a}^{-1}(\hat{g}(b(\varepsilon))) \leq \hat{g}(b(\varepsilon)) < \hat{b}(\varepsilon)$ (by Cii)
 * ($t^* \leq \bar{t}$) $\hat{b}(c(E[\|x(t^*)\|^2])) \leq m(t^*) \Rightarrow t^* \in (t_0, \bar{t}]$

③ Next, we claim $\forall T_k \in (t_0, \bar{t}]$. $t^* \neq T_k$ (by iii) - $0 < g(\epsilon) < \epsilon$.
 Proof of claim: If $t^* = T_k$, then $0 \leq \hat{b}(\epsilon) \leq m(t^*) \leq \bar{g}(m(t^*)) \leq m(t^*) \leq \hat{b}(\epsilon)$,
 which is impossible. we have $m(t) < \hat{b}(\epsilon)$, $\forall t \in [t_0 - \nu, t^*)$ (*)
 Now, we come back to prove step A:

• (no impulsive) ($t_{k-1} \leq t_k < t^* < t_d$)
 Let $\bar{t} = \sup \{t \in [t_0, t^*] \mid m(t) \leq \bar{g}(\hat{b}(\epsilon))\}$. $m(t_k) < \bar{g}(\hat{b}(\epsilon))$. (*) $t^* < t_d$
 $+ m$ continuous $\Rightarrow \bar{t} \in (t_0, t^*)$. $m(\bar{t}) = \bar{g}(\hat{b}(\epsilon))$. and $m(t) \geq \bar{g}(\hat{b}(\epsilon))$.
 $\forall t \in [\bar{t}, t^*]$. $\Rightarrow \forall s \in [-\nu, 0)$ $\bar{g}(m(t+s)) \leq \bar{g}(\hat{b}(\epsilon)) \leq m(t)$. (*)

$\Rightarrow D^+ m(t) \leq p(t) C(m(t))$, $\forall t \in [\bar{t}, t^*]$.

$$\textcircled{1} \int_{m(\bar{t})}^{m(t^*)} \frac{ds}{C(s)} \leq \int_{\bar{t}}^{t^*} p(s) ds \leq \int_{\bar{t}}^{\bar{t}+\tau} p(s) ds \leq \sup \int_{\bar{t}}^{\bar{t}+\tau} p(s) ds = M_1.$$

$$\textcircled{2} \int_{m(\bar{t})}^{m(t^*)} \frac{ds}{C(s)} = \int_{\bar{g}(\hat{b}(\epsilon))}^{\hat{b}(\epsilon)} \frac{ds}{C(s)} \geq M_2. \quad \textcircled{3} M_2 > M_1 \quad (\text{by ci v}).$$

Contradiction.

• (impulsive) ($T_k < t^* < t_{k-1}$, $k \geq 1$). $(t_0 \in [t_{k-1}, t_k])$.
 $m(t_k) \leq \hat{b}(\epsilon) \Rightarrow m(t_k) \leq \bar{g}(m(t_k)) \leq \bar{g}(\hat{b}(\epsilon))$. (by iii) $\bar{g}$ is non decreasing

Define $\bar{t} = \sup \{t \in [T_k, t^*] \mid m(t) \leq \bar{g}(\hat{b}(\epsilon))\}$. we use the.

Similar argument to get. $\bar{t} \in [T_k, t^*)$
 $m(\bar{t}) = \bar{g}(\hat{b}(\epsilon))$ $m(t) \geq \bar{g}(\hat{b}(\epsilon))$, $\forall t \in [\bar{t}, t^*)$.

Again $D^+ m(t) \leq p(t) C(m(t))$ $\forall t \in [\bar{t}, t^*)$

$$\int_{m(\bar{t})}^{m(t^*)} \frac{ds}{C(s)} \leq \int_{\bar{t}}^{t^*} p(s) ds \leq M_1$$

$$\int_{m(\bar{t})}^{m(t^*)} \frac{ds}{C(s)} = \int_{\bar{g}(\hat{b}(\epsilon))}^{\hat{b}(\epsilon)} \frac{ds}{C(s)} \geq M_2$$

$+ M_2 > M_1$, contradiction.

proof A is completed.

• Step B: From step A, we know. $\exists \eta > 0$ s.t. $E[C(t) \|x(t)\|] \leq \eta \Rightarrow E[C(t) \|x(t)\|^2] \leq \eta$.

By ci) $\Rightarrow E(V) \leq \hat{\alpha}^1(E[C(t) \|x(t)\|^2]) \leq \hat{\alpha}^1(\eta)$ $\forall t \geq t_0 - \nu$.

Let $0 < \delta < \rho_1$, and define $0 < M = m(\epsilon) = \sup \left\{ \frac{1}{C(s)} \mid s \in [\bar{g}(\hat{b}(\epsilon)), \hat{\alpha}(\rho_1)] \right\}$.

For q satisfy $\hat{b}(\epsilon) \leq q \leq \hat{\alpha}(\rho_1)$, we have $\bar{g}(\hat{b}(\epsilon)) \leq \bar{g}(q) \leq \hat{\alpha}(\rho_1)$.

$$\Rightarrow M_2 \leq \int_{\bar{g}(\epsilon)}^q \frac{ds}{C(s)} \leq M [q - \bar{g}(\epsilon)] \Rightarrow \bar{g}(\epsilon) \leq q - \frac{M_2}{M} \leq q - d.$$

where $d < (\frac{M_2 - M_1}{M}) < \frac{M_2}{M}$. $\forall \rho_1 \geq \delta > 0$, define

(R) : Th 4.2. The stabilisation ...
 ② $N = N(r) = \min \{N \in \mathbb{N} \mid \hat{\alpha}(p_i) < \delta(r) + Nd\}$
 Next, we claim, $\exists \eta > 0$, s.t. $\forall \delta > 0$, $\exists T(\delta) = T + (K+L)(N-1)$.
 $E(\|x(t)\|^2) \leq \delta$, $\forall t \geq t_0 + T(\delta)$, $t_0 \in (T_0, T_1)$

Given $0 \leq A \leq \hat{\alpha}(p_i)$, we use a similar argument in the proof of step A to show

① if $m(t) \leq A$, $\forall t \in [t_j - r, t_j]$, then $m(t+1) \leq A$, $t \geq t_j$;

② if, in addition, $\hat{\delta}(r) < A$, then $m(t+1) \leq A - d$, $\forall t \geq t_j$. ($L \leq r$).

Now, we define $k^{(i)}$ $i=1,2,\dots,N$, s.t. $T_{k^{(i)}-1} < T_{k^{(i)}} + r \leq T_{k^{(i)}}$

i.e. $\begin{cases} T_{k^{(i)}} - T_{k^{(i)-1}} \geq r \\ T_{k^{(i)}-1} - T_{k^{(i)}} < r \end{cases} \Rightarrow T_{k^{(N)}} \leq t_0 + T + (K+L)(N-1) = t_0 + T$

now, we prove step B by induction method.

For $T_{k^{(1)}} = T_1$, $A = \hat{\alpha}(p_i)$ $\Rightarrow m(t+1) \leq \hat{\alpha}(p_i) - d$.

$T_{k^{(2)}}$ $A = \hat{\alpha}(p_i) - d$ $\xrightarrow{\text{by } ② + \hat{\delta}(r) < A} m(t+1) \leq \hat{\alpha}(p_i) - 2d$

$\Rightarrow m(t+1) \leq \hat{\alpha}(p_i) - Nd \leq (\hat{\delta}(r) + Nd) - Nd = \hat{\delta}(r)$ $\forall t \geq t_0 + T$.

Through $\hat{\delta}(E(\|x(t)\|^2)) \leq m(t) \leq \hat{\delta}(r) \Rightarrow E(\|x(t)\|^2) \leq \delta$, $\forall t \geq T + t_0$ $\square$

Remark 4.1. $\xrightarrow{\text{impulsive effect.}}$

(1) Theorem 4.1 : unstable $\Rightarrow$ stable.

(2) (iv) $M_2 > M_1$, ensure that any possible growth in $\checkmark$ between impulsive is reduced by $\checkmark$ at the impulsives.

(3) $M_1 \Rightarrow \tilde{M}_1 := \sup_{k \in \mathbb{N}} \int_{T_{k-1}}^{T_k} p(s) ds$

(4) For m.s. uniform stability, $T < \infty$ is not necessary.

(5) (iii) is independent of the time delay.

(6) p th moment stability $b(\|\psi(t)\|) \leq V(t, \psi(t)) \leq a(\|\psi(t)\|)$ (Ci)
 exponential stability $a(\cdot) = a \cdot \|\psi(t)\|^p$ $b(\cdot) = b \cdot \|\cdot\|$ $c = c \cdot \|\cdot\|$

(i). For sufficient condition of stability property for system 4.1.

we need (ii) $\Rightarrow LV \leq -p(t) \in (V(t, \psi(t)))$, $V(t+s, \psi) \leq \bar{g}(V(t, \psi))$

(iv) $\Rightarrow M_2 = \sup_{t \geq 0} \int_t^{t+r} \frac{ds}{a(s)} < \inf_{t \geq 0} \int_t^{t+r} p(s) ds < \infty$

$\mu = \inf \{T_k - T_{k-1}\}$, More detail we refer to.

Corollary 4.1.

$\square$

(6): Th 4.3. The stability theorems and comparison method.
 4.2 Stability Analysis. by comparison method.
 Similar method in ODE.

Th 4.2: $\exists \alpha \in K_2$: $\liminf_{s \rightarrow \infty} \alpha(s) > 0$, and (A1), (A2).

Let $V \in Y^{1,2}([t_0, \infty) \times \mathbb{R}^n; (\mathbb{R}^+))$ satisfy
 (i) $V(t, \varphi(t)) \leq \alpha(\|\varphi(t)\|^2) \leq \alpha(\|\varphi\|_T^2)$.

(ii) $\int V' \leq h(t, V)$ $\forall t \neq T_k$. When $V(t, \varphi) \leq \alpha(\|\varphi\|_T^2)$. $\alpha \in K_3$ (non-decreasing)

h : continuous, $h(t, z)$ is concave in z .

$\lim_{(t,y) \rightarrow (T_k^-, x)} h(t,y) = h(T_k^-, x)$. exists.

(iii) $\forall T_k$, $V + \beta \leq \alpha_k(V(T_k^-, \varphi(t_0)))$, α_k is non-decreasing concave function.

(iv). the auxiliary scalar impulsive system.

$$\begin{cases} D^+ v(t) = h(t, v(t)) & t \neq T_k \\ v(t) = \alpha_k(v(t^-)) & t = T_k \\ v(t_0) = v_0 \geq 0. \end{cases}$$

has a maximal solution $v(t) = v(t, t_0, v_0)$.

Then. $E[V(t_0, x_0)] \leq v_0$ implies $E[V] \leq v(t)$. $\forall t \geq t_0$.

proof: Let $x(t)$ be any solution of system (4.1).

By Itô's formula and condition (i), we have, $\forall t \in [T_{k-1}, T_k)$

$$\begin{aligned} E[V(t)] &\leq E(V(T_{k-1})) + \int_{T_{k-1}}^t E[h(s, V(s))] ds \\ &\stackrel{=: m(t)}{\leq} E(V(T_{k-1})) + \int_{T_{k-1}}^t h(s, E(V(s))) ds. \end{aligned}$$

Jensen inequality

$$\Rightarrow D^+ m(t) \leq h(t, m(t)).$$

By (iii) $\Rightarrow m(T_k) \leq \alpha_k(m(T_k^-))$. In summary, we have

$$\begin{cases} D^+ m(t) \leq h(t, m(t)) \\ m(T_k) \leq \alpha_k(m(T_k^-)) \\ m(t_0) = E(V(t_0, x_0)). \end{cases}$$

Therefore, by standard comparison method, (see Th 1.6.1 in [2])
 we have $m(t) \leq v(t) \leq \alpha(\|x(t)\|^2)$; $\forall t \geq t_0$. □

⑥: Th 4.3. The stability properties of auxiliary scalar system (4.6)

$\Rightarrow$ stability properties of (4.1).

Proof: (i) (4.6) stability $\Rightarrow$ (4.1) is stable

We have $\forall b(\varepsilon) > 0, \exists \delta > 0$. s.t. $V_0 < \delta \Rightarrow V(t, t_0, V_0) \leq b(\varepsilon), \forall t \geq t_0$
 choose $V_0 = \alpha(\|x_0\|^2)$. $\delta_1 = \delta_1(\varepsilon) > 0$ such that $\alpha(\delta_1) < b(\varepsilon) < \delta$.

$\delta = \min\{\delta, \delta_1\}$. we claim that

$$E[\|x(t)\|^2] \leq \delta \Rightarrow E[\|x(t)\|^2] < \varepsilon, \forall t \geq t_0$$

If our claim were not true, $\exists \bar{t} \in [t_k, t_{k+1})$.

$$E[\|x(\bar{t})\|^2] \geq \varepsilon.$$

$$E[\|x(t)\|^2] < \varepsilon, \forall t \in [t_k, \bar{t}).$$

By (A1) $\Rightarrow \exists \pm$, s.t. $t_k < \pm \leq \bar{t}$ satisfying $\varepsilon \leq E[\|x(\pm)\|^2] < \rho$.

Define $m(\pm) = E(V(\pm))$. $\forall t \in [t_0, \pm]$. By Th 4.2, we get.

$$m(\pm) < r(t; t_0, \alpha(E\|x(t_0)\|^2)), \forall t \in [t_0, \pm]$$

By condition (i), we obtain

$$b(\varepsilon) \leq b(E\|x(\pm)\|^2) \leq E(b(\|x(\pm)\|^2)) \leq E(V(\pm)) = m(\pm)$$

$$b(\varepsilon) \leq m(\pm) < r(t, t_0, \alpha(E\|x(t_0)\|^2)) \leq r(t, t_0, \alpha(\delta)) < b(\varepsilon).$$

Contradiction. Therefore, we have $E[\|x(t)\|^2] < \varepsilon, \forall t \geq t_0$.

(ii) (4.6) uniformly stable $\Rightarrow$ (4.1) uniformly stable.

Choose $0 < \eta < \rho, \rho < \rho$. Assume that $\forall b(\eta) > 0, \exists \delta > 0, T = T(\eta, \delta) > 0$
 s.t. $V_0 < \delta \Rightarrow \forall t \geq t_0, V(t, t_0, V_0) < b(\eta), \forall t \geq t_0 + T$.

Similar to (i). we omit it here. $\square$.

Corollary 4.2. Let $\alpha_K V(t_k, \psi(0)) = \alpha(d_k) V(t_k, \psi(0))$.

where $d_k \geq 0$ and $d = \sum_{k=1}^{\infty} d_k < \infty$, $\alpha(d_k) \leq 1 \forall k$ ($\prod_{k=1}^{\infty} \alpha(d_k) < \infty$)

If (i) $h=0$, provided that $V(t_k, \psi(s)) \leq q(V(t, \psi(s)))$ for some $s \in [-r, 0]$, $q \in K_3$, then $x \geq 0$ of (4.1) is uniform stable in the m.s.

(ii) $h(t, V(t, \psi(s))) = -c(V(t, \psi(s)))$, q defined in (i) then $x \geq 0$ is asymptotically stable in the m.s.

Proof: Step A: V is well defined, i.e., $E[\|x(t)\|^2] \leq \rho$, for some t_0 .

Step B. Construct comparison scalar system

Step C. C.1 $t^* \in [t_{k-1}, t_k)$ no impulse? C.2 $t \in [t_k, t_{k+1})$, impulsive.

①. Let $0 < \varepsilon < \rho$, $1 \leq \bar{a} = \sum_{k=1}^{\infty} 2(d_k) < \infty$. $\delta = \delta(\varepsilon) < \bar{a}^{-1} (b(\delta)/d) \Rightarrow 0 < \delta < \varepsilon$.
 $b^{-1}(\delta) \leq \varepsilon$ $(\frac{1}{\bar{a}} < \bar{a})$.

Let $t_k \in [t_{k-1}, t_k)$ for some positive integer k and ϕ , for which $E[\|x(t_k)\|^2] \leq \delta$. We show $x(t)$ is uniformly stable, i.e. $E[\|x(t)\|^2] \leq \varepsilon$.

Step A: If not, $\exists t^*$ st. $\forall t \in [t_0 - \tau, t^*)$, $E[\|x(t)\|^2] < \varepsilon < \rho$, and either $E[\|x(t^*)\|^2] = \varepsilon$, $E[\|x(t^*)\|^2] = E(\|x_0\|^2) \neq \varepsilon$ or $E[\|x(t^*)\|^2] > \varepsilon$ $t^* = t_k$ for some k .

by (4.6) $\varepsilon < E[\|x(t^*)\|^2] < \rho$ since $E[\|x(t^*)\|^2] \leq \varepsilon < \rho$. Thus $V(t, x(t))$ is well-defined for $t \in [t_0, t^*]$.

Step B: Applying Itô formula to V and recall (i) $E(V(t)) \leq E(V(s)) + E \int_s^t L V(u, x(u)) du$. Define $m(t) = E V(t) \Rightarrow D^+ m(t) \leq E(L V(t)) \leq E C V(t) = 0$. provided $m(t+s) \leq \rho(m(t))$ (This means $m(t)$ is nonincreasing, $\forall t \in [t_0, t^*]$).

Furthermore, $\forall t_k \in [t_0, t^*]$, $m(t_k) \leq 2(d_k) m(t_k^-)$. construct comparison impulsive system $\begin{cases} D^+ V(t) < 0 & t \neq t_k \\ V(t) = 2(d_k) V(t^-) & t = t_k \\ V(t_0) = V_0 = m_0 = E[V(t_0, x_0)] \end{cases}$.

Step C: (C.1) $t_0 \in [t_{k-1}, t_k)$ $t^* \in [t_{k-1}, t_k)$. by $D^+ V(t) < 0 \Rightarrow E[V(t)] \leq V(t) \leq V_0 < \delta < \varepsilon$. $\forall t \in [t_0, t^*]$. $b(E[\|x(t)\|^2]) \leq E V(t) \leq \delta < \bar{a}^{-1} (b(\delta)/\bar{a})$. $\Rightarrow E[\|x(t)\|^2] \leq \varepsilon$ since $b^{-1}(\varepsilon) < \varepsilon$.

C.2. $t^* \in [t_k, t_{k+1})$ $k \geq 1$. $b(t) \leq 2(d_k) V(t_k^-) = 2(d_k) V(t_{k-1}) \leq 2(d_k) 2(d_{k-1}) V(t_{k-1}^-) \leq \dots \leq \bar{a} V(t_0) \leq \bar{a} \bar{a}^{-1} (b(\delta)/\bar{a}) = b(\delta)$. with the aid of Th 4.3 we have $(m(t_0) \leq \bar{a}^{-1}(\delta))$. $\bar{b}(\varepsilon) \leq \bar{b}(E[\|x(t_k)\|^2]) \leq m(t^*) \leq V(t^*) \leq \bar{a} \bar{a}^{-1}(\delta) < \bar{b}(\varepsilon)$ (definition of $\bar{b}$).

contradiction. Remark 4.2. ① (i) and (ii) $\Rightarrow D^+ E[V(t)] \leq 0$ ② (4.7) $\xrightarrow{\text{impulsive}} V \xrightarrow{\text{impulsive}} \text{stable}$

Remark 4.3. unstable $\xrightarrow{\text{impulsive}} \text{stable}$, see Corollary 4.3.

- ②. Corollary 4.3. (i) $h(t, v) = p(t) \in C(V(t))$ (4.12)
 (ii) $\exists \delta_k > 0, p_0 > 0, \forall z \in C(\mathbb{R}), \forall k \in \mathbb{N}$.

$$\int_{T_k}^{T_{k+1}} p(s) ds + \int_z^{\alpha_k(z)} \frac{ds}{c(s)} \leq -\tau_k \quad (4.13)$$

Then $x=0$ is uniformly stable in the m.s. If $\lim_{k \rightarrow \infty} \delta_k = \infty \Rightarrow x=0$ asymptotically stable.

Proof: By Th 4.2. $m(t) = E V(t)$ leads to

$$(4.14) \begin{cases} V(m(t)) \leq p(t) C(m(t)) & t \neq T_k \\ m(t) \leq \alpha_k(m(t^-)) & t = T_k \\ h(t_0) = m_0 = E(V(t_0, x_0)) \end{cases} \xrightarrow[\text{system}]{\text{auxiliary}} (4.15) \begin{cases} v(t) = p(t) C(v(t)) \\ v(t) = \alpha_k(v(t^-)) \\ v(t_0) = v_0 = m_0 \end{cases}$$

• Step A: (stable): Let $0 < \varepsilon < p_0$ and $t \in [T_1, T_2]$. Choose $\delta > 0, \delta < \min \{ \varepsilon, d(\delta) \}$ and $0 < v_0 < \delta$. We claim that $v(t) < \varepsilon, \forall t \in [t_0, T_2]$. If not, then $\exists t^* \in [t_0, T_2], \text{ s.t. } v(t^*) \geq \varepsilon$. Integrating (4.14)

$$\int_{\alpha_k(\varepsilon)}^{\varepsilon} \frac{ds}{c(s)} < \int_{v(t_0)}^{v(t^*)} \frac{ds}{c(s)} \leq \int_{t_0}^{t^*} p(s) ds \leq \int_{T_1}^{T_2} p(s) ds \quad (4.16)$$

$$\Rightarrow \int_{T_1}^{T_2} p(s) ds \leq \int_{\varepsilon}^{\alpha_k(\varepsilon)} \frac{ds}{c(s)} < 0, \text{ which contradicts (4.13)} \Rightarrow v(t) < \varepsilon, \forall t \in [t_0, T_2]$$

Next, we prove $v(t) < \varepsilon, \forall t \in [T_k, T_{k+1})$ by induction method. $(V(T_k) = \alpha_k(V(T_{k-1})))$

$$\int_{V(T_k)}^{V(t)} \frac{ds}{c(s)} = \int_{V(T_k)}^{V(t^-)} \frac{ds}{c(s)} + \int_{V(T_k)}^{\alpha_k(V(T_k^-))} \frac{ds}{c(s)} \leq \int_{T_k}^{T_{k+1}} p(s) ds + \int_{V(T_k^-)}^{\alpha_k(V(T_k^-))} \frac{ds}{c(s)} \leq -\delta_k \quad (4.17)$$

$$\Rightarrow V(t) \leq V(T_k) \Rightarrow v(t) < \varepsilon, \forall t \in [T_k, T_{k+1}). \Rightarrow v \geq 0 \text{ is uniformly stable.}$$

• Step B: (asy. stable): By (4.18) $\Rightarrow \lim_{k \rightarrow \infty} V(T_k)$ exists. We claim $\lim_{k \rightarrow \infty} V(T_k) > 0$.

If not, we assume $\lim_{k \rightarrow \infty} V(T_k) = 0 \Rightarrow \frac{V(T_{k+1}) - V(T_k)}{c(\eta)} \leq \int_{V(T_k)}^{V(T_{k+1})} \frac{ds}{c(s)} \leq -\delta_{k+1}$

where $\frac{1}{c(\eta)} = \sup \left\{ \frac{1}{c(s)} \mid \forall s \in [V(T_k), V(T_{k+1})] \right\} > 0, \lim_{k \rightarrow \infty} V(T_k) = 0$
 $\Rightarrow V(T_k) \leq V(T_{k-1}) - c(\eta) \sum_{i=1}^k \delta_i \rightarrow 0 \quad (\frac{1}{c(\eta)} > 0, \sum_{i=1}^k \delta_i \rightarrow \infty)$

$$\Rightarrow \lim_{k \rightarrow \infty} V(T_k) = 0. \quad \square$$

Conclusion of Chap. 4.

(1). Lyapunov - Razumikhin $\Rightarrow$ stability analysis
 comparison method

(2) Sect 4.1 (4.1a) stable + small impulsive $\Rightarrow$ stable

(4.1a) unstable + small impulsive frequency $\Rightarrow$ stable
 Sect 4.2. stability + unbounded impulsive $\Rightarrow$ stable Cor 4.2.

Th 4.1
 Corollary 4.3